

Introduction to Environmental Science EES102

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Toxicity

- Toxic refers to materials (pollutants) that are **poisonous** to people and other living things, typically at **ppm levels or lower**.
- Toxicity refers to the ability of a chemical to **cause harmful effects** to the body. As was described by Paracelsus (1493-1541):
 - “What is it that is not poison? All things are poison and nothing is without poison. It is the dose only that makes a thing not a poison.”
- There are a number of factors that influence the toxic effects of chemicals on the body. These include, but are not limited to:
 - The **quantity** and **concentration** of the chemical.
 - The **length of time** and the **frequency** of the exposure.
 - The **route** of the exposure.
 - If **mixtures** of chemicals are involved



Toxic Effects

Toxic effects are generally classified as **acute toxicity** or **chronic toxicity**.

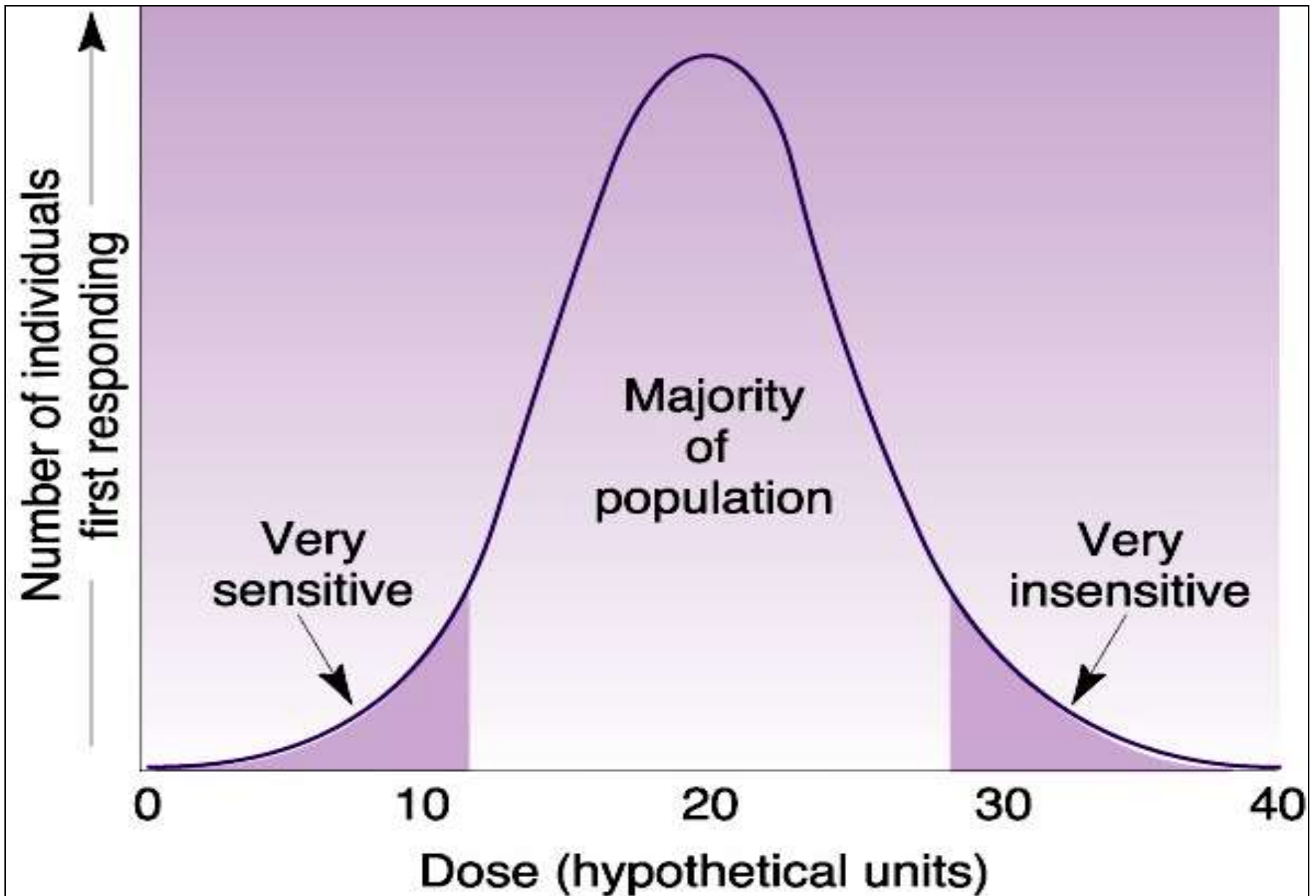
- Acute toxicity is generally thought of as a single, short-term exposure where **effects appear immediately** and are often **reversible**. An example of acute toxicity relates to the over consumption of alcohol and “hangovers”.
- Chronic toxicity is generally thought of as frequent exposures where **effects may be delayed** (even for years) and are generally **irreversible**. Chronic toxicity can also result in acute exposures, with long term chronic effects. An example of **chronic toxicity relates to cigarette smoking and lung cancer**.

Hazardous and Toxic Chemicals



- Allergens - formaldehyde
- Immune system depressants - PCBs?
- Neurotoxins - lead, DDT
- Mutagens
- Teratogens—cause birth defects, thalodimide, alcohol
- Carcinogens;
- Over 20 million organic chemicals

A Typical Dose/Response Curve



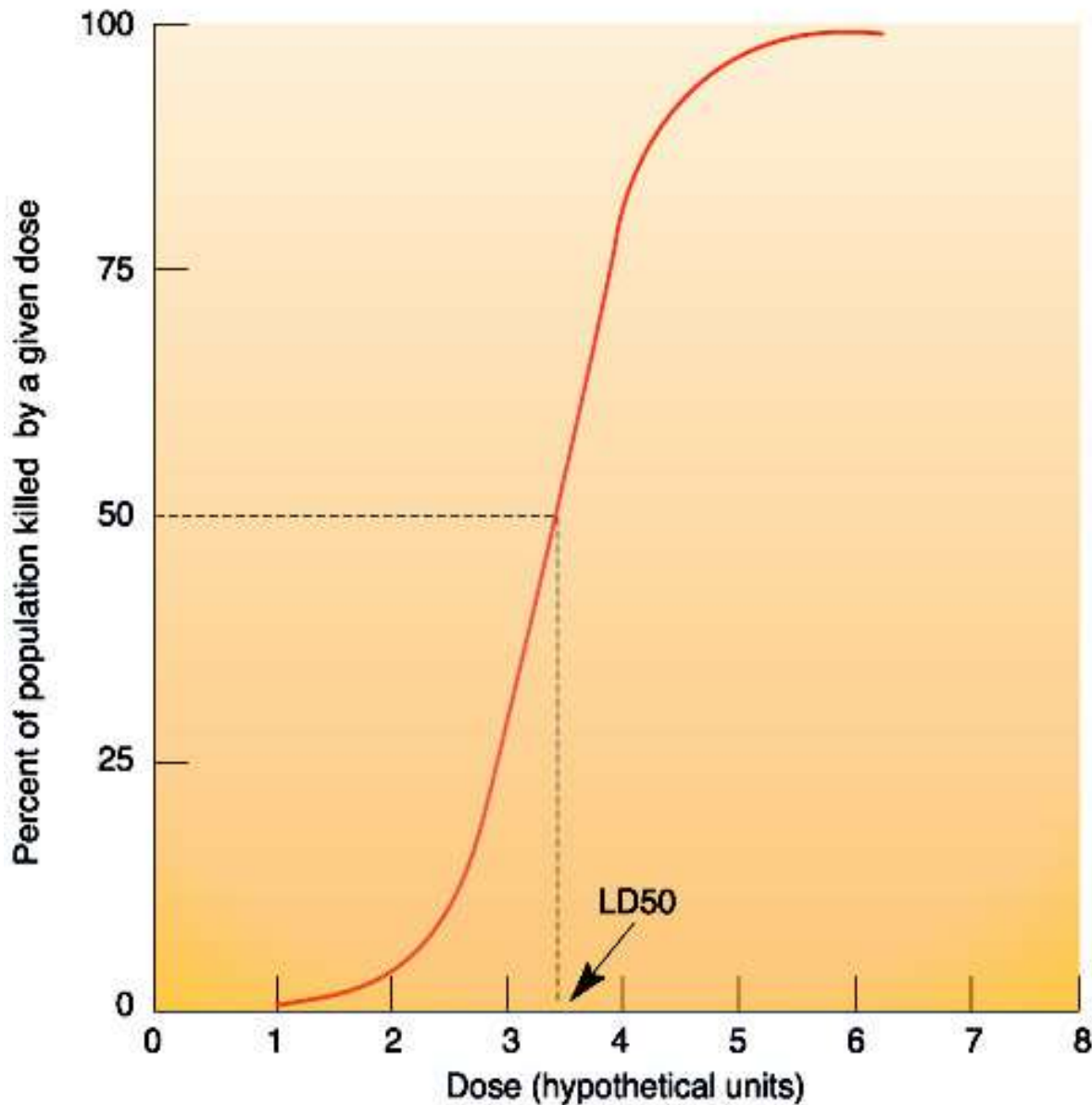
Evaluating Toxicity Data

- **SDSs** and other chemical resources generally refer to the toxicity of a chemical numerically using the term **Lethal Dose 50 (LD50)**. The LD50 describes the amount of chemical **ingested or absorbed** by the skin in test animals that **causes death in 50%** of test animals used during a toxicity test study.
- Another common term is **Lethal Concentration 50 (LC50)**, which describes the amount of chemical **inhaled by test animals** that causes death in 50% of test animals used during a toxicity test study. **The LD50 and LC50 values are then used to infer what dose is required to show a toxic effect on humans.**

Relevant Toxicity of a Chemical

Toxicity Class	Animal LD50	Probable Lethal Dose for 70 kg Person (150 lbs.)	Example
Super Toxic	Less than 5 mg/kg	A taste (7 drops or less)	Botulinum toxin
Extremely Toxic	5 - 50 mg/kg	< 1 teaspoonful	Arsenic trioxide, Strychnine
Very Toxic	50 - 500 mg/kg	< 1 ounce	Phenol, Caffeine
Moderately Toxic	0.5 - 5 g/kg	< 1 pint	Aspirin, Sodium chloride
Slightly Toxic	5 - 15 g/kg	< 1 quart	Ethyl alcohol, Acetone

In addition to having a toxic effect on the body, some chemicals can be carcinogenic, mutagenic, teratogenic, and acutely toxic.



LD50 - the dose of a toxin that is lethal to half the test population

Liver: The Detoxifier

- **Enzymes present in the liver carry out oxidation reduction reactions or coupling reactions and convert the toxins to forms that can be excreted. Thus nature has provided us with some ability to deal with toxins**
- **Why does alcohol addiction cause male impotence? Ethyl alcohol is metabolized in the liver to acetic acid and further to carbon dioxide but at the same time when more alcohol builds up in the liver, oxidizing enzyme levels increase. These same enzymes also end up oxidizing testosterone, the male hormone.**

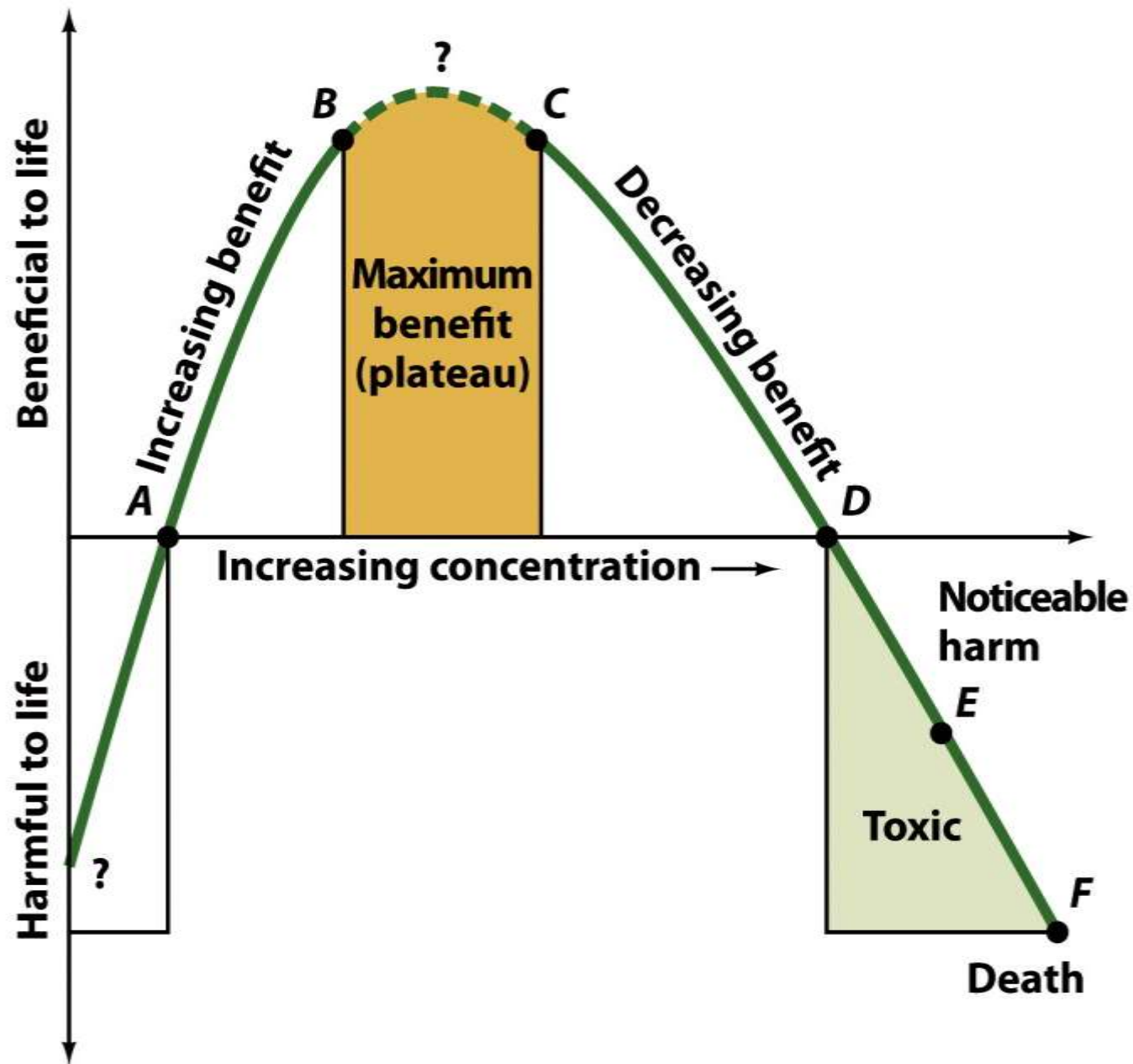
Coupling reaction: Benzene (fat soluble) is oxidized to Phenol which is then coupled to glucuronic acid and forms phenyl glucuronide (water soluble)

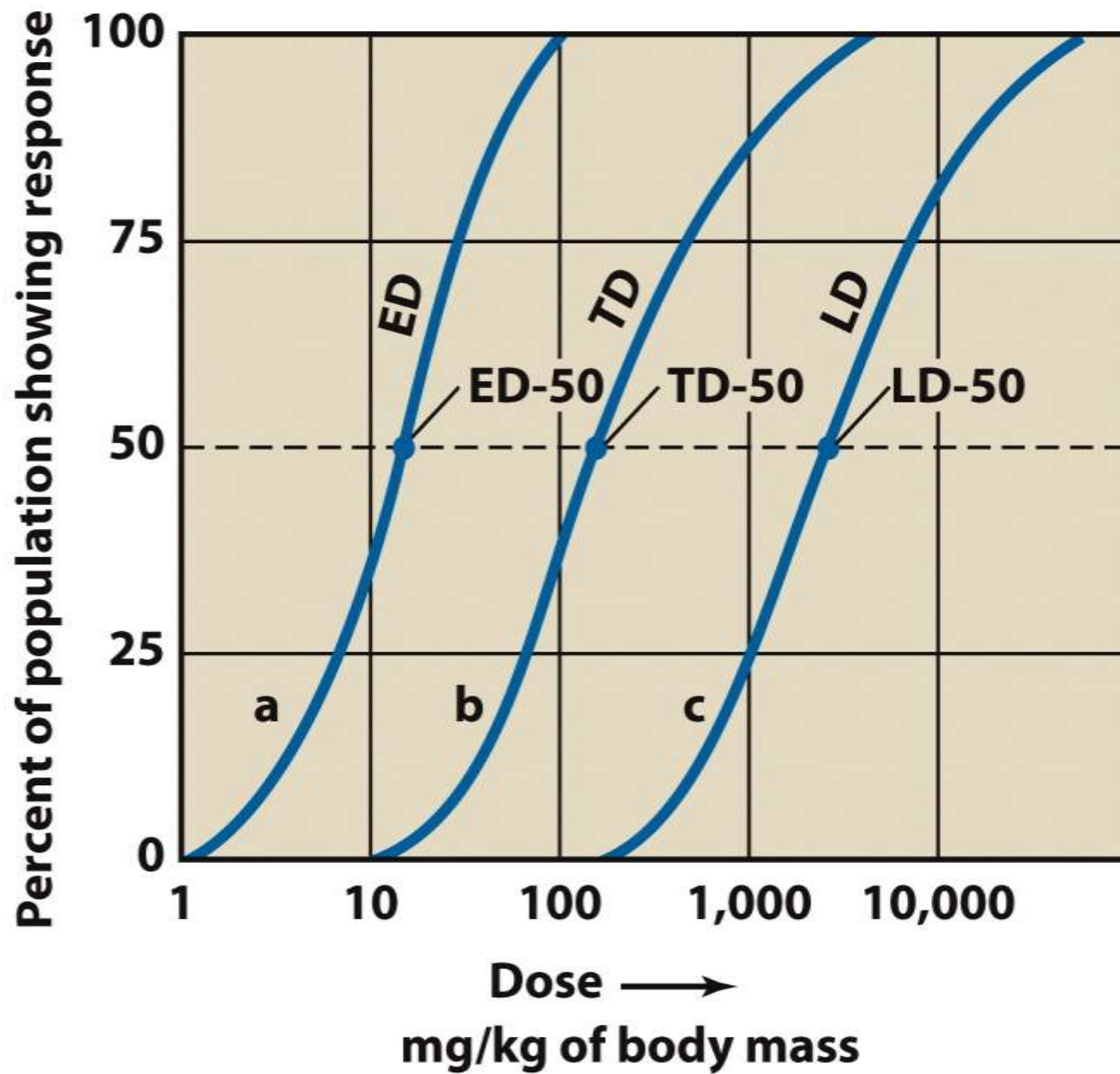
Table 15.4**Approximate LD-50 Values (for Rodents) for Selected Agents**

<i>Agent</i>	<i>LD-50(mg/kg)^a</i>
Sodium chloride (table salt)	4,000
Ferrous sulfate (to treat anemia)	1,520
2,4-D (a weed killer)	368
DDT (an insecticide)	135
Caffeine (in coffee)	127
Nicotine (in tobacco)	24
Strychnine sulfate (used to kill certain pests)	3
Botulinum toxin (in spoiled food)	0.00001

^a Milligrams per kilogram of body mass (termed mass weight, although it really isn't a weight) administered by mouth to rodents. Rodents are commonly used in such evaluations, in part because they are mammals (as we are), are small, have a short life expectancy, and their biology is well known.

Source: H. B. Schiefer, D. C. Irvine, and S. C. Buzik, *Understanding Toxicology* (New York: CRC Press, 1997).





DOSE-RESPONSE CURVE

The **ED-50 (effective dose 50%)** is the dose that causes an effect in 50% of the population of observed subjects.

E.g. ED-50 of aspirin would be the dose that relieves headaches in 50% of the people.

The **TD-50 (toxic dose 50%)** is defined as the dose that is toxic to 50% of the population.

Often used to indicate responses such as reduced enzyme activity, decreased reproductive success, or onset of specific symptoms.

E.g the TD-50 of aspirin would be 200 to 300 milligrams per kilogram of body weight symptoms are rapid or deep breathing, ringing in the ears (tinnitus), sweating, fast heart rate, hallucinations

LD-50 – (lethal dose 50%) the dose of a toxin that is lethal to half the test population.

E.g the TD-50 of aspirin would be 500 milligrams per kilogram of body weight, cause of death usually cardiovascular failure

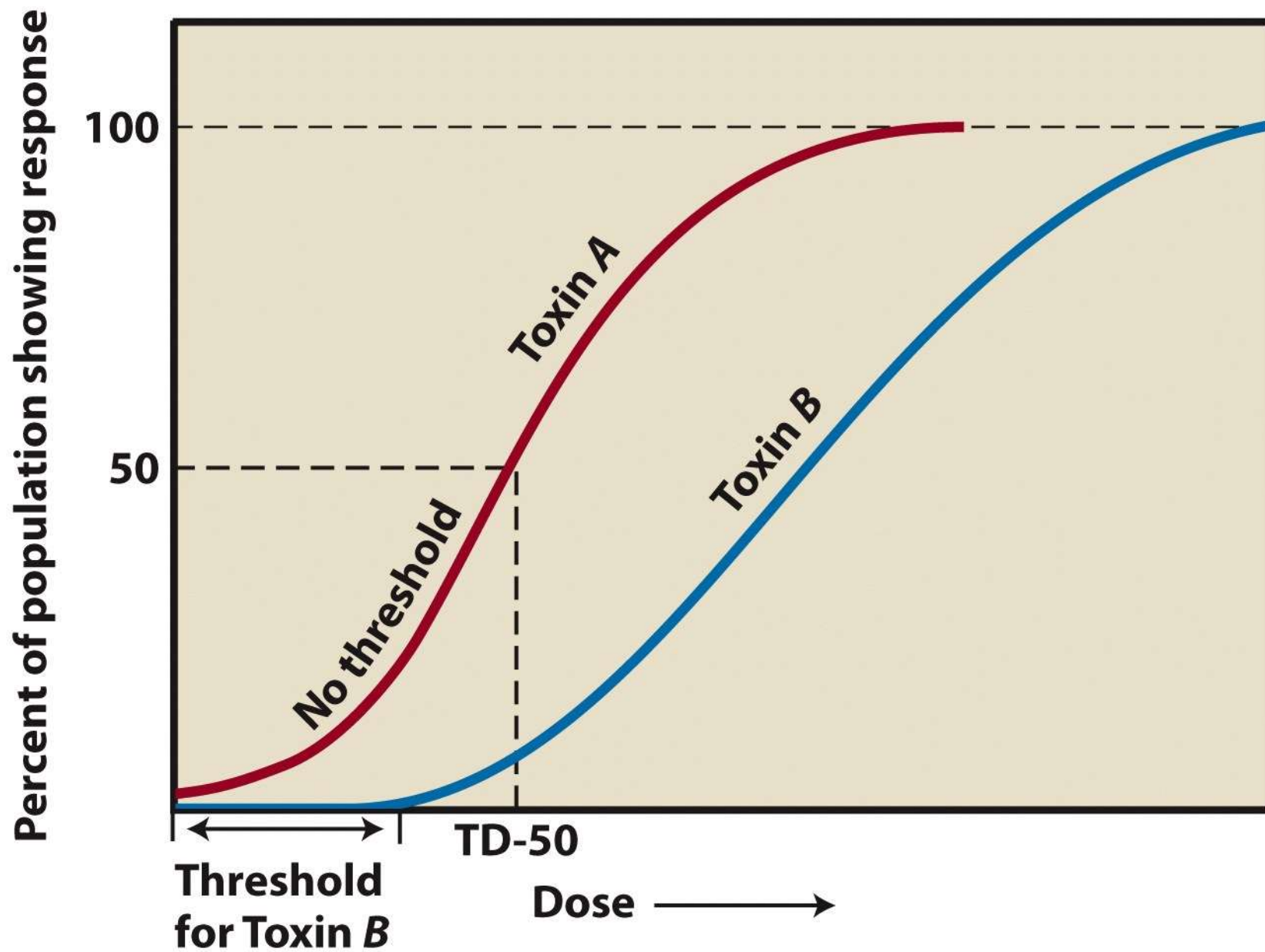
The therapeutic index is defined as the ratio of the LD-50 to the ED-50. The greater the therapeutic index, the safer the drug.

THRESHOLD EFFECTS

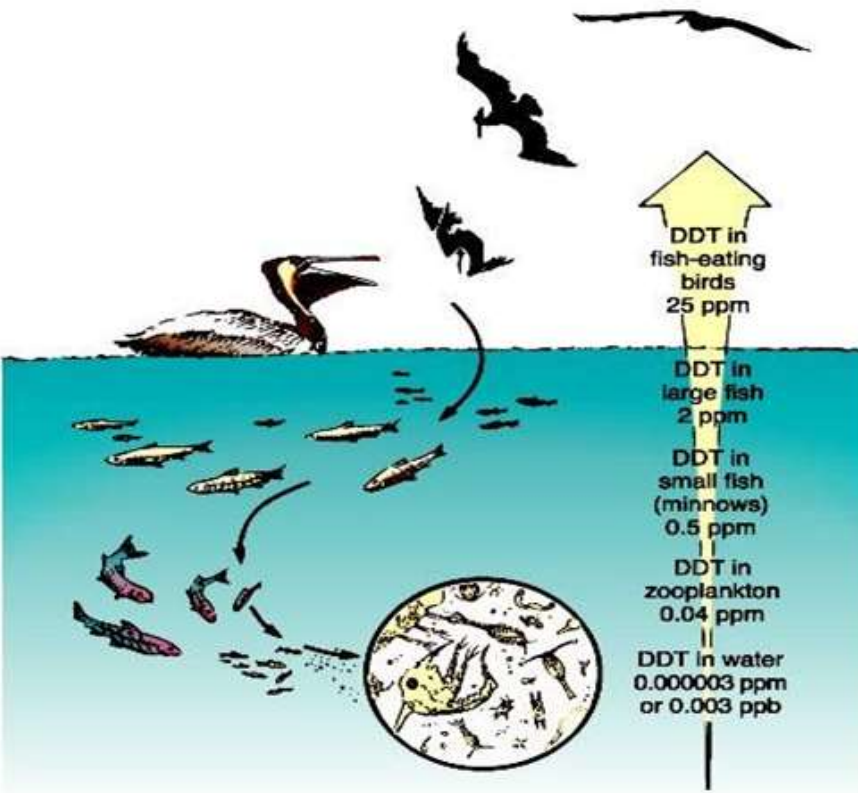
Threshold is a level below which no effect occurs and above which effects begin to occur.

- If a threshold exists, then a concentration below the threshold is safe.
- If there is no threshold dose, then even the smallest amount has some negative toxic effect.

A problem in evaluating thresholds for toxic pollutants is that it is difficult to account for synergistic effects.



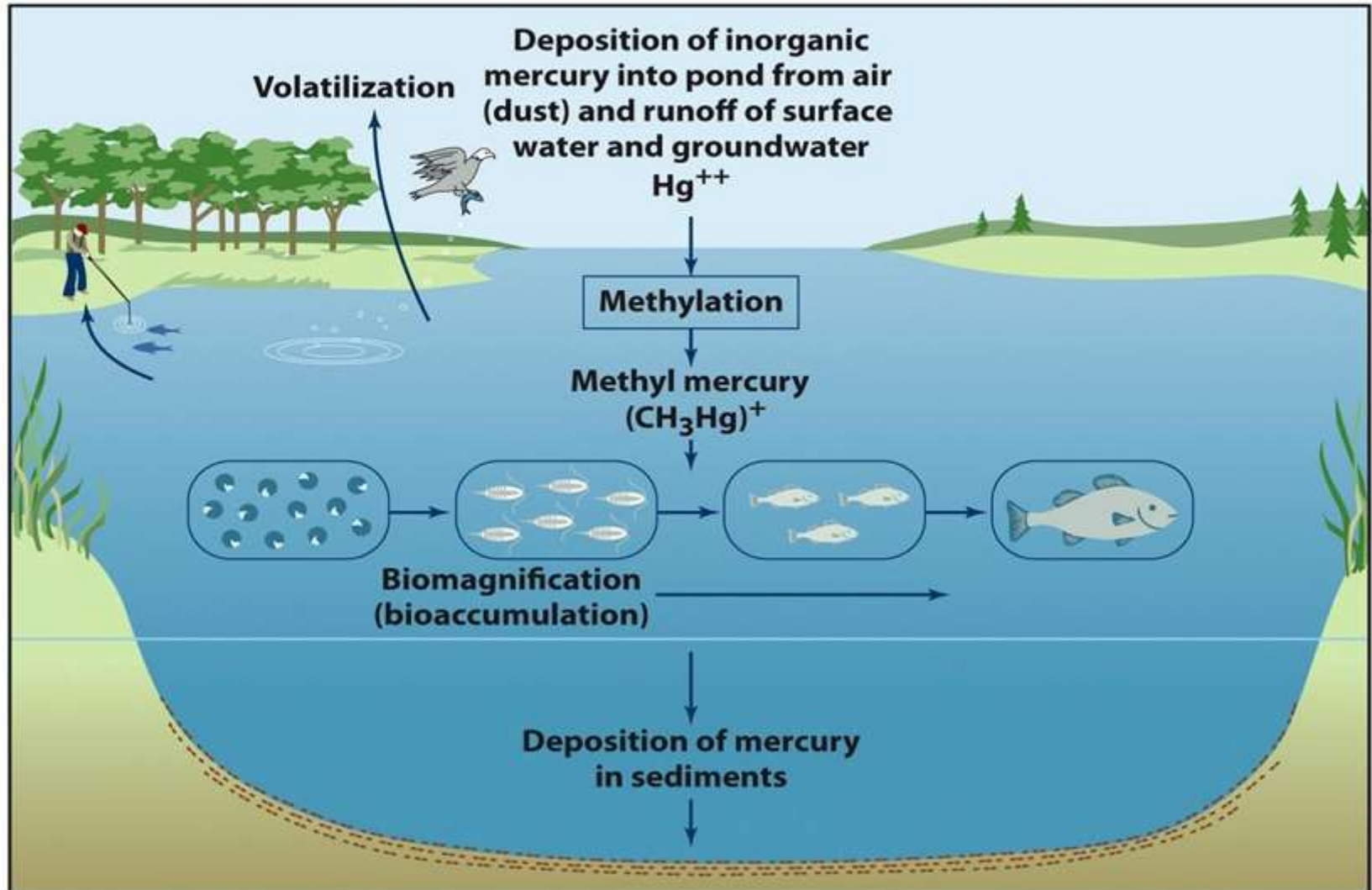
Bioconcentration, Bioaccumulation and Biomagnification



- **Bioaccumulation** - diluted toxins in the environment can reach dangerous levels inside cells and tissues
- **Biomagnification** - the effects of toxins are magnified through food webs

Example of Bioaccumulation: Minamata

The Minamata Convention (10 October 2013, Geneva)



What Can Hit Us Where?

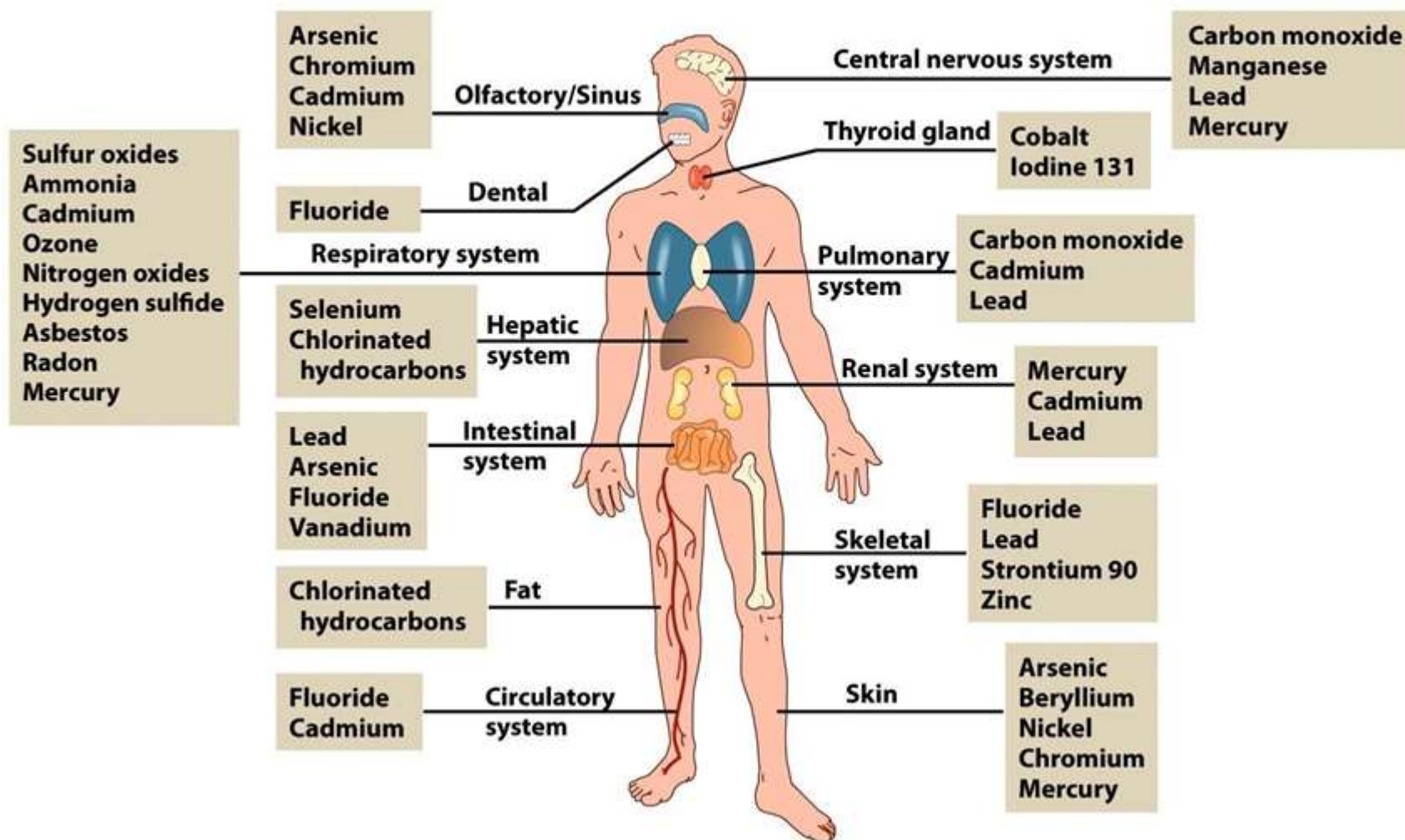


Table 15.3**Effects of Pollutants on Wildlife**

<i>Effect on Population</i>	<i>Examples of Pollutants</i>
Changes in abundance	Arsenic, asbestos, cadmium, fluoride, hydrogen sulfide, nitrogen oxides, particulates, sulfur oxides, vanadium, POPs ^a
Changes in distribution	Fluoride, particulates, sulfur oxides, POPs
Changes in birth rates	Arsenic, lead, POPs
Changes in death rates	Arsenic, asbestos, beryllium, boron, cadmium, fluoride, hydrogen sulfide, lead, particulates, selenium, sulfur oxides, POPs
Changes in growth rates	Boron, fluoride, hydrochloric acid, lead, nitrogen oxides, sulfur oxides, POPs

^a Pesticides, PCBs, hormonally active agents, dioxin, and DDT are examples (see Table 15.1).

Source: J. R. Newman, *Effects of Air Emissions on Wildlife*, U.S. Fish and Wildlife Service, 1980. Biological Services Program, National Power Plant Team, FWS/OBS-80/40, U.S. Fish and Wildlife Service, Washington, D.C.

POPs

POPs may produce a hazard for decades or hundreds of years.

First produced when their harm was not known

Now banned or restricted

Table 15.1

Selected Common Persistent Organic Pollutants (POPs)

<i>Chemical</i>	<i>Example of Use</i>
Aldrin ^a	Insecticide
Atrazine ^b	Herbicide
DDT ^a	Insecticide
Dieldrin ^a	Insecticide
Endrin ^c	Insecticide
PCBs ^a	Liquid insulators in electric transformers
Dioxins	By-product of herbicide production

Source: Data in part from Anne Platt McGinn, "Phasing Out Persistent Organic Pollutants," in Lester R. Brown et al., *State of the World 2000* (New York: Norton, 2000).

^a Banned in the United States and many other countries.

^b Degrades in the environment. It is persistent when reapplied often.

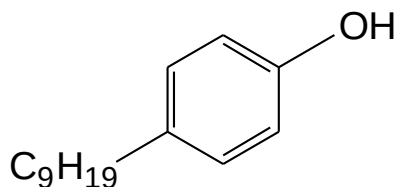
^c Restricted or banned in many countries.

"DDT - Powerful Insecticide, Harmless to Humans"

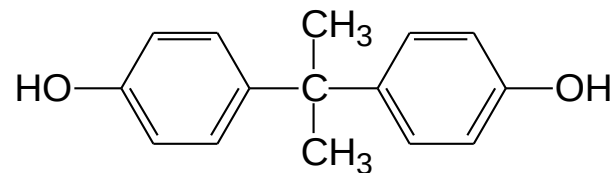


Structures of Some POPs

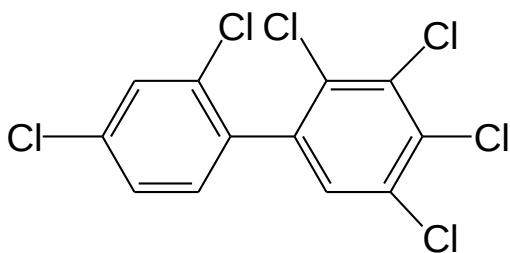
Additive derived chemicals



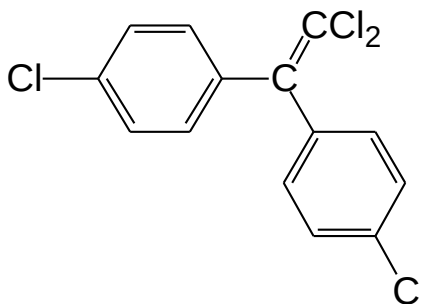
Nonylphenol



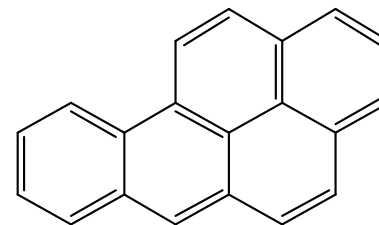
Bisphenol A



**Polychlorinated biphenyl
(PCBs)**



DDTs



**Polycyclic aromatic
hydrocarbons (PAHs)**

PERSISTENT ORGANIC POLLUTANTS

POPs have several properties that define them:

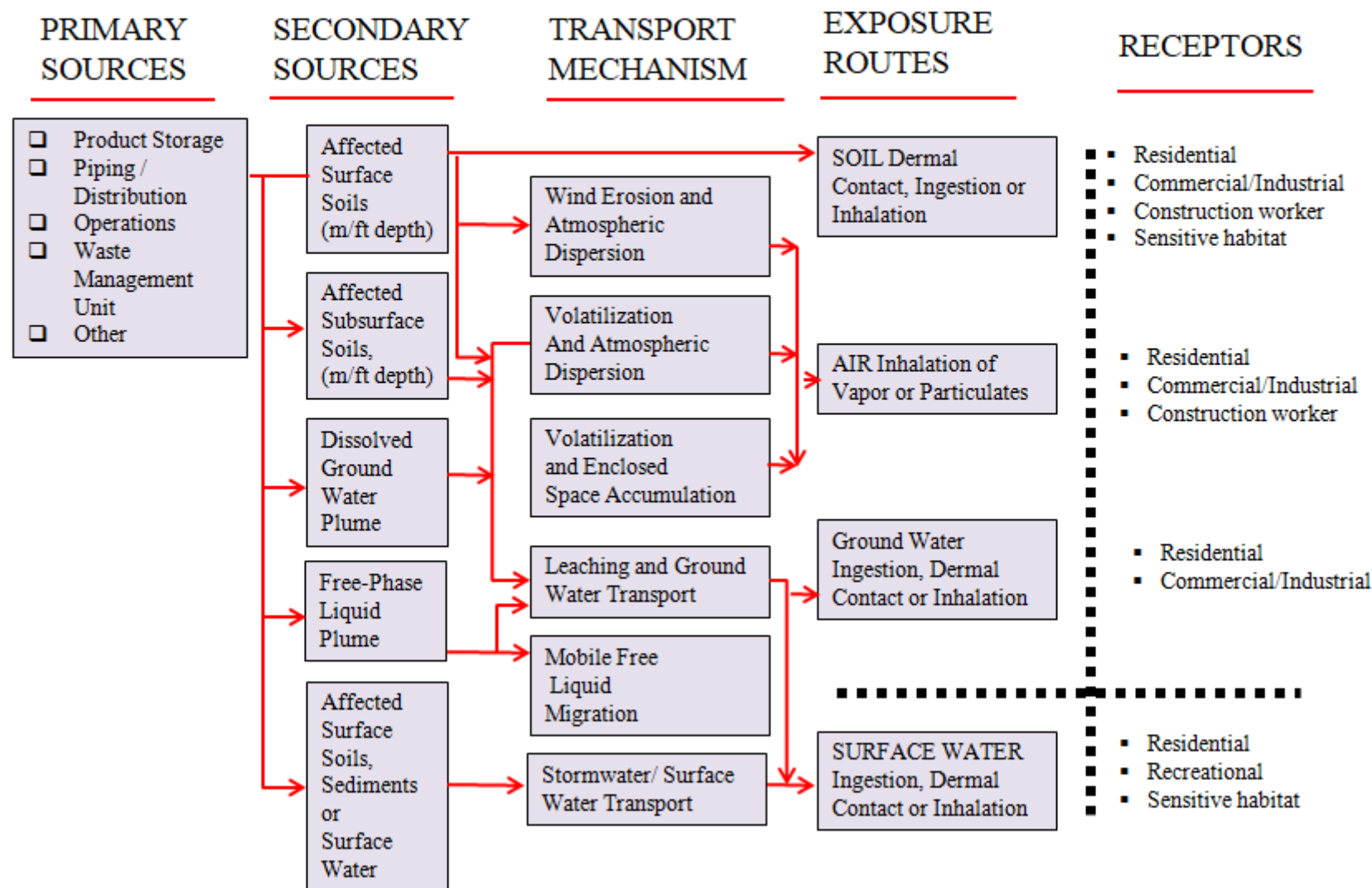
- They have a carbon-based molecular structure, often containing highly reactive chlorine.
- Reactive chlorine attacks cell walls of microbes and destroys the cells
- Most are synthetic chemicals.
- They do not easily break down in the environment.

RISK ASSESSMENT

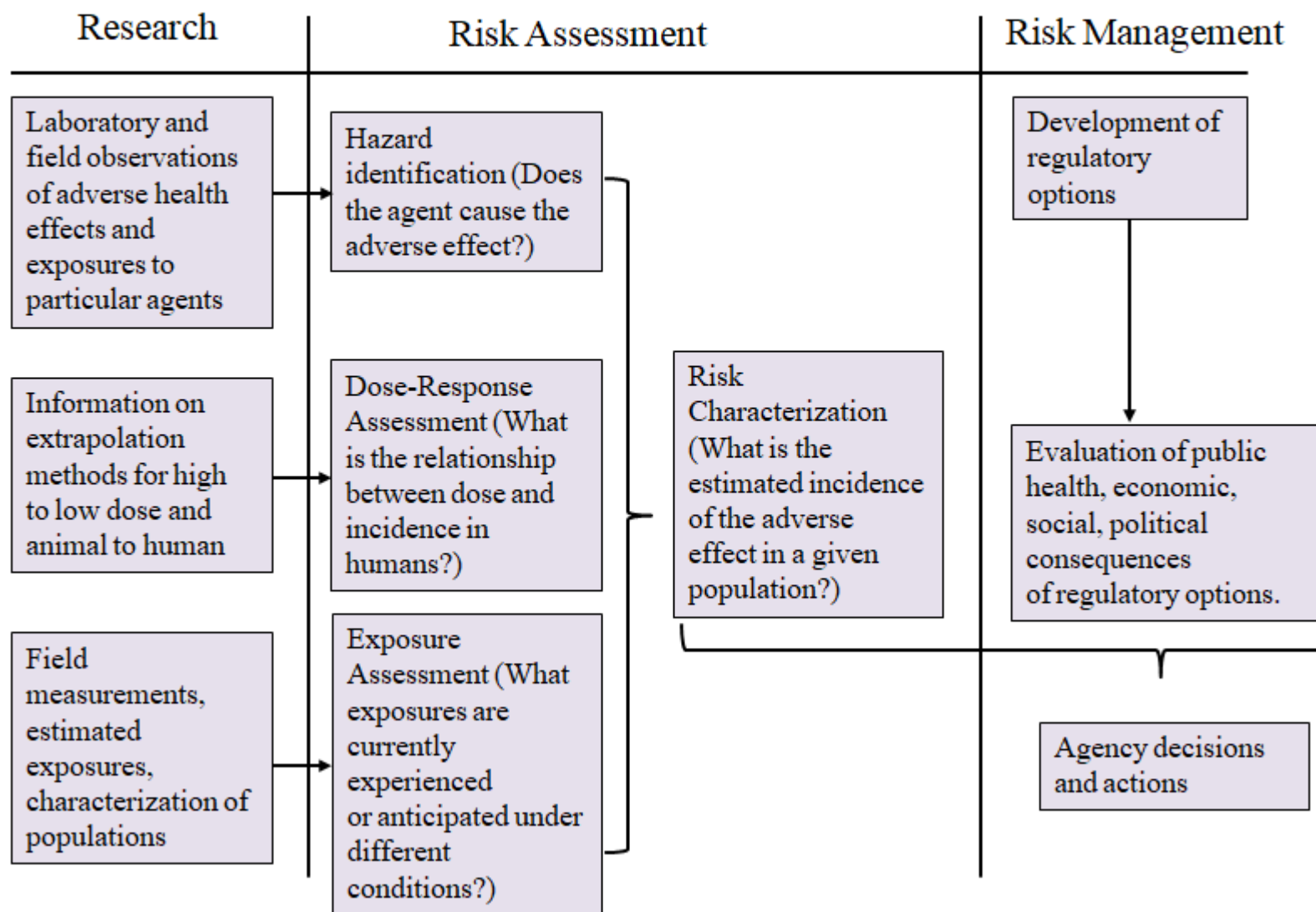


$$\text{Risk} = \text{Toxicity} \times \text{Exposure}$$

Exposure Pathways Evaluation



Federal Risk Assessment Paradigm



RISK ASSESSMENT

Such an assessment generally includes four steps:

- Identification of the hazard.
- Dose–response assessment.
- Exposure assessment.
- Risk characterization.

Risk assessment is difficult, costly, and controversial.

Risk management integrates the assessment of risk with technical, legal, political, social, and economic issues.

Risk Characterization

Quantifying cancer risks

estimated as the incremental probability of the occurrence of cancer over a lifetime as a result of exposure to the potential carcinogen

$$IELCR = \text{Cancer Slope Factor} \times \text{Lifetime Average Daily Dose}$$

IELCR = Incremental Excess Lifetime Cancer Risk

Exposure Assessment

- Example of an Intake Equation for Ingestion of Drinking Water

$$\text{Intake (mg/kg-day)} = \frac{C_{\text{mg}} \times \text{IR} \times \text{EF} \times \text{ED}}{\text{BW} \times \text{AT}}$$

C_{mg} = Concentration in water of chemical of concern (mg/l)

IR = Ingestion Rate (l/day)

EF = Exposure Frequency (days/yr)

ED = Exposure Duration (yr)

BW = Body Weight (kg)

AT = Averaging Time (period over which exposure is averaged) (days)

For noncarcinogens: AT = ED * 365 days per year and intake is called Chronic Daily Intake (CDI).

For carcinogens: AT = Lifetime (70 years) * 365 days per year and intake is called Lifetime Average Daily Dose (LADD).

Example Intake Equations

Ingestion of Drinking Water

C_w = Concentration of chemical(s) of concern in water (mg/L)

$$\text{Intake (mg/kg-day)} = \frac{C_w \times IR \times EF \times ED}{BW \times AT}$$

Inhalation of Air

C_A = Concentration of chemical(s) of concern in air (mg/m³)

$$\text{Intake (mg/kg-day)} = \frac{C_A \times IR \times EF \times ED}{BW \times AT}$$

Ingestion of Soil

C_s = Concentration of chemical(s) of concern in soil (mg/kg)

CF = Conversion Factor (10⁻⁶ kg/mg)

$$\text{Intake (mg/kg-day)} = \frac{C_s \times IR \times CF \times EF \times ED}{BW \times AT}$$